

Project Summary

The **Spotify Data Analysis SQL Project** is a comprehensive data analytics portfolio project designed to explore, clean, and extract meaningful business insights from a Spotify dataset. The end-to-end journey encompassed downloading raw data from Kaggle, performing data inspection and cleaning (such as handling anomalies like zero-duration tracks), and importing the structured dataset into a PostgreSQL database.

To address real-world analytical scenarios, the project formulated and solved 15 structured business questions divided into three distinct difficulty tiers (Easy, Intermediate, and Advanced) using advanced SQL queries.

Detailed Insights & Business Analysis

1. Easy Category (Data Retrieval & Basic Aggregation)

- **Track Popularity Milestone (Q1):** Filtering tracks with over 1 billion streams helps identify blockbuster hits and top-tier global performers. **Insight:** This highlights the platform's heavy hitters that drive the vast majority of user engagement and royalties.
- **Artist & Album Mapping (Q2):** Listing distinct albums with their respective artists maps out the catalog's structure. **Insight:** Essential for inventory management and understanding artist discography distributions.
- **Licensed Content Engagement (Q3):** Aggregating comments for tracks where licensed = TRUE measures community interaction on official content. **Insight:** Licensed tracks typically receive higher engagement, reflecting stronger marketing or official distribution support.
- **Album Format Analysis (Q4):** Isolating tracks belonging to the 'single' album type. **Insight:** Helps analyze modern release trends where artists frequently drop singles rather than full-length albums to capture continuous listener attention.

- **Artist Track Count (Q5):** Counting total tracks per artist. **Insight:** Identifies prolific creators versus artists with smaller, highly curated catalogs.

2. Intermediate Category (Filtering, Grouping & Multi-Condition Logic)

- **Album Danceability (Q6):** Calculating average danceability per album. **Insight:** Helps categorize albums by mood or genre compatibility, useful for curating automated workout or party playlists.
- **Peak Energy Tracks (Q7):** Identifying the top 5 tracks with the highest energy values. **Insight:** Pinpoints high-intensity songs that dominate energetic or high-tempo listener segments.
- **Official Video Performance (Q8):** Evaluating views and likes for tracks with official videos. **Insight:** Quantifies the multiplier effect of visual media on fan appreciation and track reach.
- **Album-Level View Distribution (Q9):** Summing total views for tracks associated with each album. **Insight:** Reveals whether success is driven by breakout hit singles or uniform popularity across the whole album.
- **Platform Preference Comparison (Q10):** Comparing streams on Spotify versus YouTube using conditional logic (CASE WHEN). **Insight:** Highlights platform-specific audience behaviors—identifying tracks that overperform on video-heavy platforms compared to audio-streaming platforms.

3. Advanced Category (Window Functions & Subqueries)

- **Artist Top Tracks Ranking (Q11):** Using DENSE_RANK() partitioned by artist to find the top 3 most-viewed tracks per creator. **Insight:** Discovers each artist's signature flagship songs versus their deeper, lesser-streamed album tracks.

- **Liveness Benchmark (Q12):** Filtering tracks with a liveness score above the overall dataset average using a subquery. **Insight:** Isolates live-recording performances or tracks with a live-concert feel, appealing to niche audiences seeking authentic acoustic experiences.
- **Energy Variance per Album (Q13):** Finding the difference between highest and lowest energy values in each album using a WITH clause (CTE). **Insight:** Measures stylistic consistency; albums with low energy variance maintain a steady mood, while high variance indicates dynamic shifts.
- **Energy-to-Liveness Ratio (Q14):** Filtering specialized metric ratios greater than 1.2. **Insight:** Pinpoints studio-produced tracks that combine high drive with controlled acoustic environments.
- **Cumulative Engagement (Q15):** Calculating the cumulative sum of likes ordered by views using window functions. **Insight:** Visualizes how user appreciation compounds across the platform as overall viewership scales.

Project Outcome

This project significantly improved my ability to work with SQL in a real-world business context. Instead of focusing only on SQL syntax, I learned how to transform raw data into meaningful business insights by following a complete analytical workflow. From understanding the dataset and building a relational database to performing exploratory analysis, solving business problems, and documenting the entire process, this project strengthened my practical SQL skills and analytical thinking. It also provided valuable experience in presenting a professional, portfolio-ready data analytics project using industry-standard tools and best practices.